Part II Astrophysics: Michaelmas term

Published course schedules. Source: https://www.ast.cam.ac.uk/current-students/undergraduate/part-ii/teaching-and-learning/michaelmas-term-courses
Retrieved 15 October 2026.

Q. Introduction to Astrophysics (non-examinable)

1. Basics: Stellar magnitudes, parallax, use of distance modulus and absolute magnitude, period luminosity relationship for Cepheid variables. Hubble's law, using HR diagrams to estimate distances. Stellar collapse time, Kelvin Helmholtz timescales, the virial theorem, nuclear fusion, stellar evolution, endpoints of stellar evolution. Stars on the main sequence. [2]

2. Binary stars, Kepler's law, tides, Planck function, Stefan Boltzmann law, electromagnetic radiation, absorption cross section and optical depth. Continuum radiation types, Compton and inverse Compton scattering. Bremsstrahlung, synchrotron radiation, synchrotron self-absorption, atomic spectra, 21 cm radiation, molecular spectra, the Doppler effect, interstellar absorption, line widths and profiles, luminosity classes and P Cygni profiles. [2]

3. Telescopes, instruments and their limitations, adaptive optics, interferometers and signal-to-noise calculations. General review of the current range of instruments astronomers can access both on the ground and in space. Optical and radio telescopes, ultrahigh energy cosmic ray detection. Astronomical instrumentation: detectors including CCD and CMOS detectors. Quantum dot technology and quanta image sensors. Multi-object spectroscopy including fibre fed instruments. Diffraction limit and the need for adaptive optics, earth rotation synthesis and the measurement and importance of signal-to-noise in astronomy. [2]

4. White dwarfs, neutron stars and pulsars. Discovery of white dwarfs, degeneracy pressure, mass radius relation for white dwarfs, as they cool, the approach to becoming a black hole: pulsars and their discovery. Pulsar properties and understanding their magnetic field structure. Llife and death of pulsars, pulsars as magnetic dipoles. Using pulsars as precision clocks, gravitational wave detection. Binary stars, supernovae and accretion physics. Measuring properties of binary stars, different kinds of binaries, supernovae in binary systems, gravitational potential in binary systems, Roche lobes, accretion and the Eddington limit. Accretion onto magnetic stars, accretion with significant angular momentum, cataclysmic variables, x-ray boosters and stellar mass black holes. [4]

5. Supernovae, gamma ray bursts, SNR evolution. Supernovae types, nuclear burning, core collapse, neutrino detection, importance of P Cygni profiles, light curves, evolution of supernovae remnants. Discovery of gamma ray bursts, GRB characteristics, evidence for their great distance, the data from BATSE. SWIFT data, different kinds of gamma-ray bursts, the fireball shock model, evidence for nearby SNRs in the past. [4]

6. Active Galactic Nuclei. Early discoveries, Quasar discovery, importance of high redshifts, Quasar spectra, AGN energetics, mass accretion rate, host galaxy properties, constraints on Quasar models. Reverberation mapping , beamed radiation in AGN jets, the black holes at the centre of M 87 and possibly the Milky Way. Super luminal expansion, unified model of quasars and their energetics. Models of Quasar evolution, evidence for black holes in relatively normal galaxies, optical and radio rotation curves, dynamics of stars in the centre of the Milky Way, the black hole in our galaxy. [2]

7. Galaxies and clusters of galaxies. Galaxy types and general characteristics. Galaxy populations, interactions and collisions. Rich clusters of galaxies and their properties, interaction between galaxies and intergalactic matter and magnetic fields. Hot gas in the centres of clusters, the importance of multi-

wavelength imaging , rich clusters of galaxies and their use as cosmological telescopes. Justification for the belief in dark matter, cluster masses from x-ray observations. The Sunyaev-Zeldovich effect. Cooling flows within galaxies and clusters. Importance of weak shocks. Galaxy correlations, including Tully-Fisher relation. [2]

8. Gravitational Lensing. General principles of gravitational lensing, Einstein rings, caustics and critical lines for elliptical mass distribution. Rich clusters of galaxies and their lensing, measuring the Hubble constant, microlensing of exoplanets. [2]

9. Exoplanets. Detection of exoplanets, statistics of those found. Information we can extract from a variety of different eclipsing events. Importance of dust in forming exoplanets. Observational support evidence, forces on dust grains. The Hill radius. The difficulty in detecting earth like planets. Plans for future detection programs. [2]

10. Magnetic fields in astronomy. Magnetic fields in our Sun, space weather and dangers from it. Measuring magnetic fields, synchrotron radiation, Faraday rotation, Zeeman splitting, equi partition considerations, celebration cosmic rays and the creation of magnetic fields. Extra/intra-galactic magnetic fields. [1]

11. Observational Cosmology. Big bang observations, Hubble's law, nuclide abundances, cosmic microwave background, redshifts surveys, CMB experiments, radio surveys, infrared results, dark matter properties, cosmic neutrinos, gravitational waves and background. [2]

Recommended Reading

- Carroll, B.W. & Ostlie, D.A. An Introduction to Modern Astrophysics (Addison-Wesley) 2017

Q. Stellar Dynamics and Structure of Galaxies

1. Introduction to gravitating systems in the Universe. [1]
   Scope and motivation of stellar and galactic dynamics. Galaxies as composite systems of stars, gas, and dark matter; why dynamics is needed to recover the mass distribution and reconstruct formation histories. Collisionless versus collisional components; stellar filling factor and why galaxies can often be treated as collisionless systems whereas gas cannot. Observational overview of simple hot stellar systems, including globular clusters, open clusters, and galaxy clusters. Surface brightness, characteristic radii, velocity dispersion, crossing time, mass-to-light ratio, and the first dynamical evidence for dark matter from galaxy clusters. Introduction to the main tools of galactic dynamics: gravitational potential, density, orbits, observables, and phase-space distribution functions.

2. Orbits in a given potential [4]
   Newtonian gravity and gravitational potential; relation between force, potential, and density. Motion in a central potential; conservation of energy and angular momentum; proof that motion under a radial force law is planar. Equations of motion in plane polar coordinates and the orbit equation. Keplerian motion under the inverse-square law: ellipses, conic sections, bound and unbound trajectories, turning points, and the interpretation of orbital energy. Effective potential, circular orbits, and escape velocity. Kepler's laws and applications to the Galactic Centre and compact objects. Two-body and binary-star dynamics, including centre-of-mass motion, reduced mass, orbital angular momentum, and selected mechanisms of orbital evolution such as mass loss and gravitational radiation. General radial force laws, radial and azimuthal periods, apsidal precession, and worked examples beyond the Kepler problem.

3. Derivation of potential from density distribution [2]
   Derivation and use of Poisson's equation for self-gravitating systems. Integral form of the potential, Green-function approach, and the role of the divergence theorem and Gauss's theorem. From observed structure to dynamical model: deprojection ideas, galaxy morphology as context, and the relation

between projected and three-dimensional density profiles. Potentials and forces for spherical mass distributions, including the homogeneous sphere, modified Hubble profile, and power-law density models. Circular-speed and escape-speed curves for spherical systems, and the connection between density slope and orbital behaviour.

4. Circular and nearly circular orbits [2]
   Circular motion in a fixed potential and the relation between angular frequency and the local gradient of the potential. Stability of circular orbits from radial perturbations and the effective potential. Small-amplitude radial oscillations, epicyclic frequency, and the epicyclic approximation. Apsidal precession and the dependence of precession rate on the form of the potential. Extension to more general axisymmetric potentials: circular motion in the mid-plane, vertical perturbations about the plane, vertical oscillation frequency, and the interpretation of near-circular disk orbits in galaxies.

5. Axisymmetric density distributions and disk dynamics [5]
   Gravitational potentials generated by flattened and axisymmetric matter distributions. Why spherical systems are mathematically simpler than disks; Poisson's equation in axisymmetry; Legendre-series solutions and the potential of a ring. Thin-disk potentials and the use of Bessel functions in solving Laplace's and Poisson's equations. Circular velocity in thin disks and explicit examples including Mestel, exponential, and Kuzmin disks. Rotation curves of disk galaxies and of the Milky Way. Oort constants from local stellar kinematics, their relation to shear, vorticity, and epicyclic motion, and modern measurements using Gaia data. Observational rotation curves from H I 21 cm emission, the flat outer rotation curves of spirals, the baryonic Tully-Fisher relation, and the dynamical case for dark matter halos.

6. Collisionless systems, Jeans equations and virial methods [4]
   Meaning and domain of validity of the collisionless approximation. Two-body relaxation, crossing and relaxation times, estimates for globular clusters and galaxies, and the transition between collisional and collisionless behaviour. Gravitational drag, focusing, and dynamical friction. The distribution function in phase space and the collisionless Boltzmann equation, including its form in cylindrical polar coordinates and its relation to continuity of phase-space flow. Jeans equations as velocity moments of the Boltzmann equation and their analogy with fluid equations. Applications to isotropic, spherical, and axisymmetric systems, including asymmetric drift, the local mass density in the Solar neighbourhood (the Oort limit), spherical mass estimators, and the mass-anisotropy-density degeneracy. Tensor and scalar virial theorems and their use in estimating masses of stellar systems.

7. Jeans theorem and distribution functions [3]
   Integrals of motion and the construction of steady-state stellar systems. Jeans theorem and its implications for equilibrium distribution functions. Building self-consistent models from a chosen potential-density pair. Eddington inversion for isotropic spherical systems and conditions for physical distribution functions. Worked examples of distribution functions in simple potentials, including the harmonic oscillator and the homogeneous sphere. Power-law and double-power-law density models, polytropic distribution functions, the Plummer model, and singular and non-singular isothermal spheres. Interpretation of these models in terms of density structure, velocity dispersion, and the connection between theoretical distribution functions and observable stellar systems.

8. Globular clusters: observations, models, and evolution [3]
   Observed properties of globular clusters: ages from isochrones, age-metallicity trends, Galactic distribution, masses, tidal radii, and their contribution to early star formation and galaxy assembly. Tidal truncation, stripping, and tidal tails as signatures of interaction with the host galaxy. King models as lowered isothermal distribution functions; solution of the associated Poisson equation; core, scale, and tidal radii; projected density and velocity-dispersion profiles. Dynamical evolution of globular clusters through relaxation, evaporation and escape of stars, core collapse, binary heating, and mass segregation.

Systems with anisotropic velocity distributions and extensions beyond isotropic King models, including Michie-type models. Use of globular clusters as laboratories for stellar dynamics and as tracers of galaxy formation.

Recommended Reading

- † Binney, J. & Tremaine, S.D. Galactic Dynamics, Princeton University Press (2008).
- Landau & Lifshitz Mechanics, Pergamon (3rd edition 1976, reprinted 1994).
- Goldstein Classical Mechanics, Addison-Wesley (2nd edition 1980).
- † Binney, J. & Merrifield, M. Galactic Astronomy, Princeton University Press (1998).
- Sparke, L.D. & Gallagher, J.S. Galaxies in the Universe - An Introduction CUP (2000) (ISBN 0-521-59740-4)

Q. Structure and Evolution of Stars

1. Basic Concepts and Observational Properties: Course overview; Mass, Temperature, Luminosity Gravity, composition, Age; Photometry and stellar colours; Spectra and spectral lines;

2. Distance: Parallax, apparent and absolute magnitudes; Masses from binary stars;

3. Temperature: Black-body radiation, Wien’s Law; The Hertzsprung-Russell Diagram and spectral classification

4. Stellar Structure: Timescales; dynamical, thermal nuclear. Energy generation, thermonuclear reactions. Energy transport; opacity, radiative and convective transport. Equations of stellar structure. Hydrostatic equilibrium, Virial Theorem, Pressure. Stellar properties as a function of mass, homology. Degeneracy: Chandrasekar limit.

5. Stellar Evolution and the Hertzsprung-Russell diagram: Pre-main sequence evolution, Hayashi and Henyey tracks. Post-main sequence evolution: massive stars, supernovae, neutron stars, black holes. Post-main sequence evolution: low-mass stars, planetary nebulae, white dwarfs, Type Ia supernovae. Initial mass function

6. Observational Tests and Constraints: The mass-luminosity relationship. Stellar abundances. The most massive stars and stellar winds. Supernovae

Recommended Reading

- Prialnik, D., An Introduction to the Theory of Stellar Structure and Evolution, Cambridge University Press, 2000, 2nd Edition 2010
- Lamers, H J G L M and Levesque E M, Understanding Stellar Evolution, IOP Publishing, 2017
- Guidry, M, Stars and Stellar Processes, Cambridge University Press, 2019
- LeBlanc, F, An Introduction to Stellar Astrophysics, Wiley, 2010

Q. Relativity (Shared with Physics)

1. Introduction: problems with Newtonian gravity, the equivalence principle, gravity as spacetime curvature.

2. Foundations of special relativity: Inertial frames, spacetime geometry, Lorentz transformations, length contraction and time dilation, Minkowski line element, particle worldlines and proper time, Doppler effect, addition of velocities, acceleration and event horizons in special relativity.

3. Manifolds, coordinates and tensors: Concept of a manifold, curves and surfaces, coordinate transformations, Riemannian geometry, intrinsic and extrinsic geometry, the metric tensor, lengths and volumes, local Cartesian coordinates, pseudo- Riemannian geometry, scalar fields, vectors and dual vectors, tensor fields, tensor algebra, covariant differentiation and the metric connection, intrinsic derivative, parallel transport, geodesics.

4. Minkowski space and particle dynamics: Cartesian inertial coordinates, Lorentz transformations, 4-tensors and inertial bases, 4-vectors and the lightcone, 4-velocity, 4-acceleration, 4-momentum of massive and massless particles, relativistic mechanics, arbitrary coordinate systems.

5. Electromagnetism: Lorentz force, the current 4-vector, the electromagnetic field tensor and field equations, the electromagnetic 4-potential.

6. Spacetime curvature: Locally-inertial coordinates, weak gravitational fields, intrinsic curvature, the curvature tensor, the Ricci tensor, parallel transport, geodesic deviation and tidal effects, physical laws in curved spacetime.

7. Gravitational field equations: the energy-momentum tensor, perfect fluids, relativistic fluid dynamics, the Einstein equations, the weak field limit, the cosmological constant.

8. Schwarzschild spacetime: static isotropic metrics, solution of empty-space field equations, Birkhoff's theorem, gravitational redshift, trajectories of massive particles and photons. Singularities, radially-infalling particles, event horizons, Eddington- Finkelstein coordinates, gravitational collapse, tidal forces.

9. Experimental tests of general relativity: precession of planetary orbits, the bending of light.

10. Friedmann–Robertson–Walker spacetime: the cosmological principle, comoving coordinates, the maximally-symmetric 3-space, the FRW metric, geodesics, cosmological redshift, the cosmological field equations.

11. Linearised gravity and gravitational waves: weak field metric, linearised field equations, Lorenz gauge, wave solutions of linearised field equations.*

**The topics starred in the schedules may be lectured, but questions will not be set on them in examinations.*

Recommended Reading

- M. P. Hobson, G. P. Efstathiou and A. N. Lasenby General Relativity: An Introduction for Physicists, CUP 2006

Q. Principles of Quantum Mechanics (Shared with Mathematics)

1. Dirac formalism: Bra and ket notation, operators and observables, probability amplitudes, expectation values, complete commuting sets of operators, unitary operators. Schrodinger equation, wave functions in position and momentum space. [3]

2. Time evolution operator: Schrodinger & Heisenberg pictures, Heisenberg equations of motion. [2]

3. Harmonic oscillator: Analysis using annihilation, creation and number operators. Significance for normal modes in physical examples. [2]

4. Multiparticle systems: Composite systems and tensor products, wave functions for multiparticle systems. Symmetry or antisymmetry of states for identical particles, Bose and Fermi statistics, Pauli exclusion principle. [3]

5. Perturbation theory: Time-independent theory; second order without degeneracy, first order with degeneracy. [2]

6. Angular momentum: Analysis of states ljm> from commutation relations. Addition of angular momenta, calculation of Clebsch-Gordan coefficients. Spin, Pauli matrices, singlet and triplet combinations for two spin half states. [4]

7. Translations and rotations: Unitary operators corresponding to spatial translations, momenta as generators, conservation of momentum and translational invariance. Corresponding discussion for rotations. Reactions, parity, intrinsic parity. [3]

8. Time-dependent perturbation theory: Interaction picture. First-order transition probability, the golden rule for transition rates. Application to atomic transitions, selection rules based on angular momentum and parity, absorption, stimulated and spontaneous emission of photons. [3]

9. Quantum basics: Quantum data, qubits, no cloning theorem. Entanglement, pure and mixed states, density matrix. Classical determinism versus quantum probability, Bell inequality for singlet two-electron state, GHZ state. [2]

Recommended Reading

- E. Merzbacher Quantum Mechanics, 3rd edition. Wiley 1998
- B.H. Bransden and C.J. Joachain Quantum Mechanics, 2nd edition. Pearson
- J. Binney and D. Skinner The Physics of Quantum Mechanics. Cappella Archive, 3rd edition
- P.A.M. Dirac The Principles of Quantum Mechanics. Oxford University Press 1967, reprinted 2003
- S. Weinberg Lectures on Quantum Mechanics. CUP, 2nd ed., 2015
- J.J. Sakurai and J.J. Napolitano Modern Quantum Mechanics. CUP 2017

Further information about this course is available on the Department of Mathematics course pages.

Examples papers are available on the DAMTP Examples page.

Part II Astrophysics: Lent term

Published course schedules. Source: https://www.ast.cam.ac.uk/current-students/undergraduate/part-ii/teaching-and-learning/lent-term-courses
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Q. Topics in Astrophysics

Most of your Part II Astrophysics courses follow a strictly disciplinary approach: teaching specific physics to tackle specific astrophysical problems. Yet, much research is inherently interdisciplinary, motivated primarily by outstanding questions about the Universe and then adopting whatever mix of methods, observational and theoretical, are required to address these questions. In Topics, the aim is to develop your familiarity with this more research-oriented, question-first, approach to astrophysics, by exploring interdisciplinary astrophysics via two different, but complementary approaches: applying specific physical concepts to diverse astrophysical phenomena; and, applying diverse physical concepts to specific research themes, focusing on the physics of tides and the evolutionary pathway from interstellar gas to the formation of planets.

In the first 3 lectures we present some of the tools that are required in order to get an approximate grasp on problems where all the information is not necessarily available. This is largely an exercise in sorting out the important from the unimportant and requires a good understanding of timescales and distributions. During the remainder of the first half of the course we investigate how a specific physical concept (that of tides) can be applied to understanding a wide diversity of astrophysical phenomena from quasars and supermassive black holes, to stellar clusters, asteroids and moons. This exemplifies how mastery of specific physical concepts can open up a diverse range of astrophysical processes to investigation.

In the second half of the course, we focus on the evolutionary pathway between star forming gas and dust in the interstellar medium to the creation of protoplanetary discs. We look at the processes involved in setting up the conditions where planets (both those observed in the solar system and in the thousands of recently discovered extrasolar planetary systems) can be formed. Throughout you will be learning how to break problems down into elements that can be tackled by simple calculations using the physics that you already know or will have learned elsewhere in Part II Astrophysics.

To further your understanding of astrophysics as a research discipline, the core lectures from the course are interspersed with `Guest Lectures', in which a number of staff in astronomy departments across Cambridge will provide a perspective on their cutting-edge research. Lecture slots will also be devoted to a small number of Example Class type sessions where the main problem solving skills will be applied and discussed to sample problems. The dates of the Guest Lectures and of the Example Classes within the Sections below will be announced.

Section 1: Introduction to problem solving techniques [4]

Section 2: The physics of tides and its astronomical applications on all scales [8]

Section 3: From molecular cloud cores to discs to the brink of planet formation [12]

Recommended Reading

There are no textbooks to support the course. However, for students who are interested in learning more about some of the topics covered, the following books (in addition to those supporting other Part II Astro. courses) are recommended:

P. Armitage, The Astrophysics of Planet Formation, Cambridge University Press, 2010.
Frank, J., King, A., Raine, D., Accretion Power in Astrophysics, Cambridge University Press, 2002
D. Ward-Thompson & A. Whitworth, An Introduction to Star Formation, Cambridge University Press, 2011

Q. Astrophysical Fluid Dynamics (shared with Physics)

Fluids are ubiquitous in the Universe on all scales. As well as obvious fluids, (e.g. the gas that is in stars or clouds in the interstellar medium) a variety of other systems are amenable to a fluid dynamical description - including the dust that makes up the rings of Saturn and even the orbits of stars in the galactic potential. Although some of the techniques of conventional (terrestrial) fluid dynamics are relevant to astrophysical fluids, there are some important differences: astronomical objects are often self-gravitating or else may be accelerated by powerful gravitational fields to highly supersonic velocities. In the latter case, the flows are highly compressible and strong shock fronts are often observed (for example, the spiral shocks that are so prominent in the gas of galaxies like the Milky Way).

In this course, we consider a wide range of topical issues in astronomy, such as the propagation of supernova shock waves through the interstellar medium, the internal structure of stars and the variety of instabilities that affect interstellar/intergalactic gas. These include, perhaps most importantly, the Jeans instability whose action is responsible for the formation of every star and galaxy in the Universe. We also deal with exotic astronomical environments, such as the orbiting discs of gas which feed black holes.

1. Introduction: The concept of a fluid. Collisional and collisionless fluids. Kinematics. Conservation of mass. Pressure. (Inviscid) momentum equation for a fluid under gravity. Stress tensor and the concept of ram pressure. [2]
2. Basic Concepts of Gravity: Poisson's equation. Gravitational potential. The Virial Theorem. [1]
3. Equation of State: Barotropic relation between pressure and density. Energy equation. Hydrostatic equilibrium. Examples: hydrostatic atmosphere under uniform gravity; self-gravitating isothermal slab; self-gravitating polytropes as simple models of stars, mass-radius relation. [4]
4. Sound Waves: Sound speed: adiabatic and isothermal case. Sound waves in a stratified atmosphere. [1]
5. Supersonic Flows: Rankine-Hugoniot conditions for adiabatic and isothermal shocks. Application to blast waves and supernova remnants. [3]
6. Bernoulli's Equation and its Applicability: De Laval nozzle and its relevance to astrophysical jets. Bondi accretion, stellar winds and mass loss. [2]
7. Fluid Instabilities: Convective instability, Schwarzschild criterion. Jeans instability. Rayleigh-Taylor and Kelvin-Helmholtz instability. Thermal instability, Field criterion. [3]
8. Viscous Flows: Linear shear flow. Navier-Stokes equation. Vorticity and energy dissipation in viscous flows. Accretion discs. Steady thin discs. [3]
9. Magnetohydrodynamics: The ideal MHD equations. Alfven waves. [3]
10. Computational Astrophysical Fluid Dynamics [2]

Recommended Reading

- Clarke, C.J. & Carswell, R.F. Principles of Astrophysical Fluid Dynamics, Cambridge University Press (2014) Landau & Lifshitz, Fluid Mechanics, Pergamon Press (1987)

Further Suggestions

- Acheson, D. Elementary Fluid Dynamics Oxford University Press (1990)
- Batchelor, G.K. An Introduction to Fluid Dynamics, Cambridge University Press (1967, reprinted 2000) Lamb, H. Hydrodynamics, Cambridge University Press (6th ed 1932, reprinted 1993

Q. Introduction to Cosmology

1. Is Cosmology Science?: Causal structure of space-time: our past light cone. Technology horizon, particle horizon and the `size' of the Universe. Importance of symmetry principles in cosmology. Big problems in cosmology. [1]
2. The Background Cosmology: Symmetric spaces. The Friedmann-Robertson-Walker metric. Energy Momentum Tensor for Perfect Fluid. Friedman equations and geometry of the Universe. Cosmological redshift. Newtonian cosmology. Cosmological constant. de-Sitter space and time-slicing. Horizons. Distances and age of the Universe.[6]
3. Thermal History: The cosmic microwave background (CMB) radiation. Thermal equilibrium: bosons and fermions. Particle content at early times. Neutrinos and neutrino decoupling. Big-bang nucleosynthesis. Relic particles. Dark matter. Baryon asymmetry. Recombination. [5]
4. Fluctuations: Newtonian perturbation theory. Fluctuations in the CMB – Silk Damping. Inflation and the origin of fluctuations. Gravitational waves and tests of inflation. What can we learn from the CMB? The LCDM model. The multiverse. Unanswered questions. [5]
5. Observational Probes: Standard candles – Type 1a supernovae. Standard ruler – Baryon Acoustic Oscillations. Inverse distance ladder and H0. Forward distance ladder and H0. Gravitational lensing. Redshift space distortions. Is LCDM consistent with observations? [4]
6. Non-Linear Evolution of Structure: Spherical collapse model. The mass function. Galaxy formation. [4]

Recommended Reading

- Modern Cosmology, Dodelson D., Schmidt F., 2020, Academic Press
- Principles of Physical Cosmology, Peebles P.J.E., 1993, Princeton University Press
- General Relativity: An Introduction for Physicists, Hobson M.P., Efstathiou G. and Lasenby A.N., 2006, Cambridge University Press
- Introduction to Cosmology, Ryden B., 2016 (2nd edition)
- Cosmology, Bauman D., 2022

Q. Statistical Physics (shared with DAMTP)

Further information about this course is available on the Department of Mathematics course pages.

Examples papers are available on the DAMTP Examples page.

1. Fundamentals of Statistical Mechanics: Microcanonical ensemble. Entropy, temperature and pressure. Laws of thermodynamics. Example of paramagnetism. Boltzmann distribution and canonical ensemble. Partition function. Free energy. Specific heats. Chemical potential. Grand Canonical Ensemble. [5]
2. Classical Gases: Density of states and the classical limit. Ideal gas. Maxwell distribution. Equipartition of energy. Diatomic gas. Interacting gases. Virial expansion. Van der Waals equation of state. Basic kinetic theory. [3]
3. Quantum Gases: Density of states. Planck distribution and black body radiation. Debye model of phonons in solids. Bose-Einstein distribution. Ideal Bose gas and Bose-Einstein condensation. Fermi-Dirac distribution. Ideal Fermi gas. Pauli paramagnetism. [8]
4. Thermodynamics: Thermodynamic temperature scale. Heat and work. Carnot cycle. Applications of laws of thermodynamics. Thermodynamic potentials. Maxwell relations. [4]
5. Phase Transitions: Liquid-gas transitions. Critical point and critical exponents. Ising model. Mean field theory. First and second order phase transitions. Symmetries and order parameters. [4]

Recommended Reading

- F. Mandl Statistical Physics. Wiley 1988
- R.K. Pathria Statistical Mechanics, 2nd ed. Butterworth-Heinemann 1996

- L.D. Landau and E.M. Lifshitz Statistical Physics, Part 1 (Course of Theoretical Physics volume 5).
- M. Kardar Statistical Physics of Particles. CUP 2007
- F. Reif Fundamentals of Thermal and Statistical Physics. McGraw-Hill 1965
- A.B. Pippard Elements of Classical Thermodynamics. Cambridge University Press, 1957
- K. Huang Introduction to Statistical Physics. Taylor and Francis 2001